

Data Structures



Lecture 16 Tree (Basics)

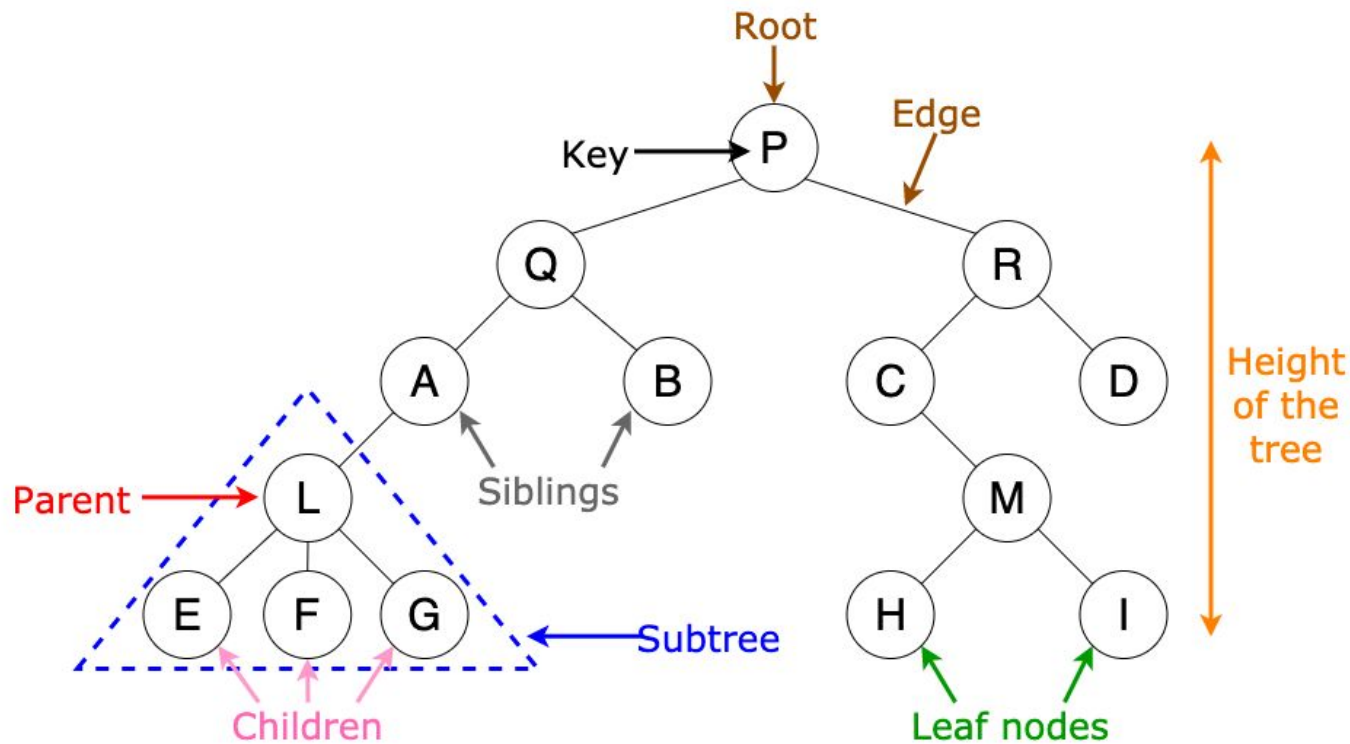
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Trees



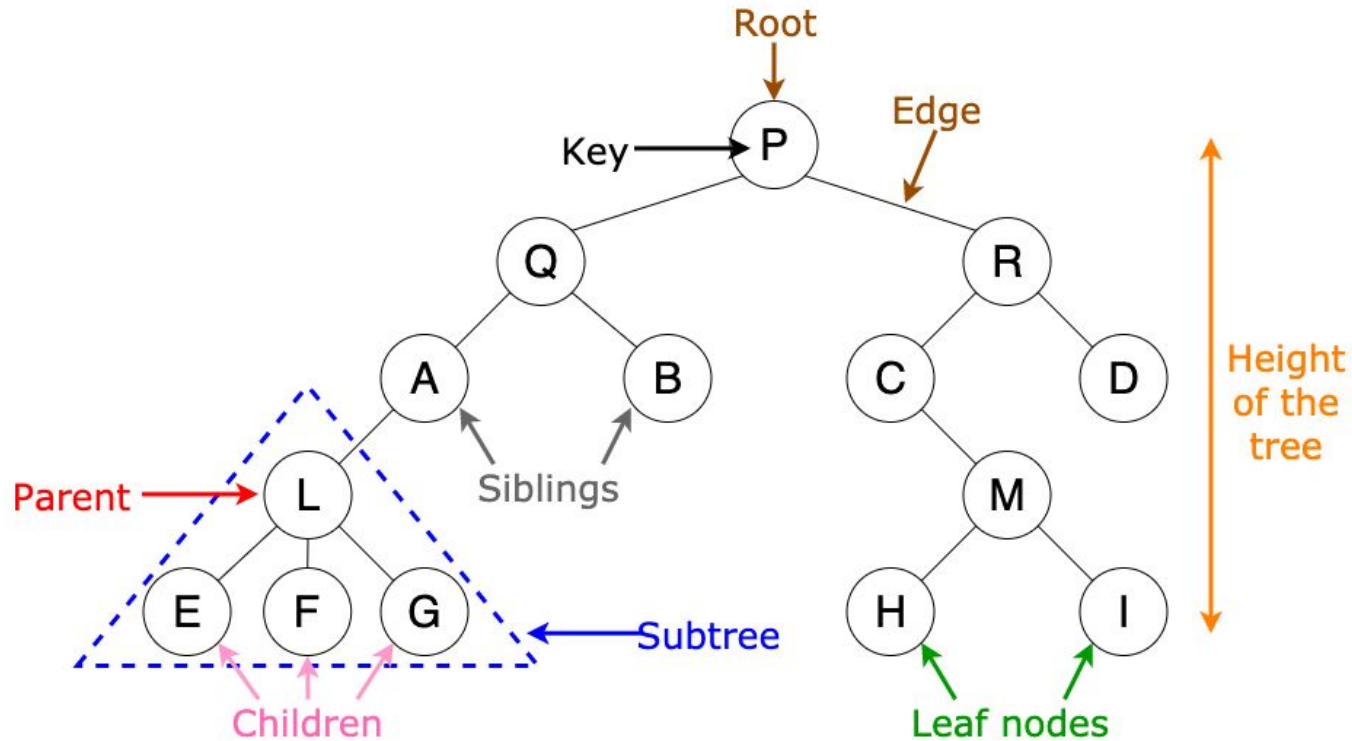
Why Trees?

Sorting New Elements

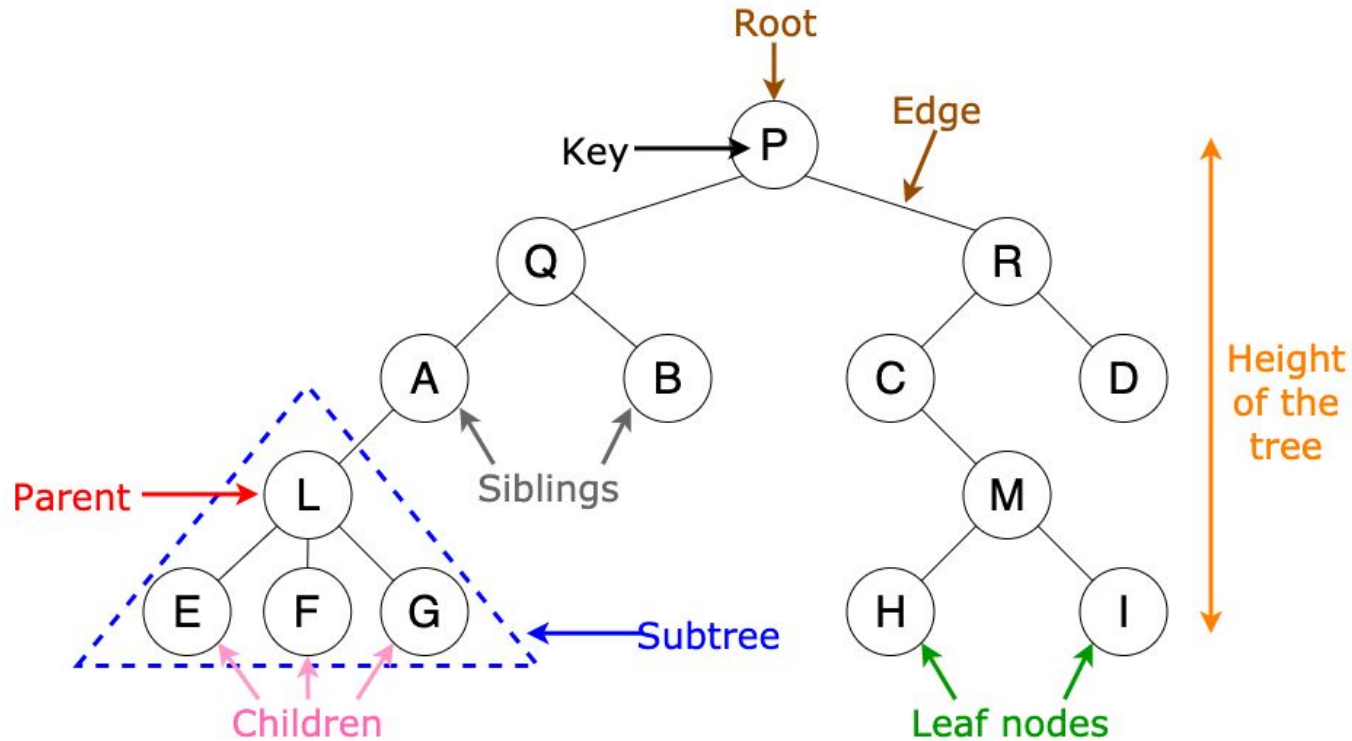
Folder/File System Structure

Computer Network Algorithms

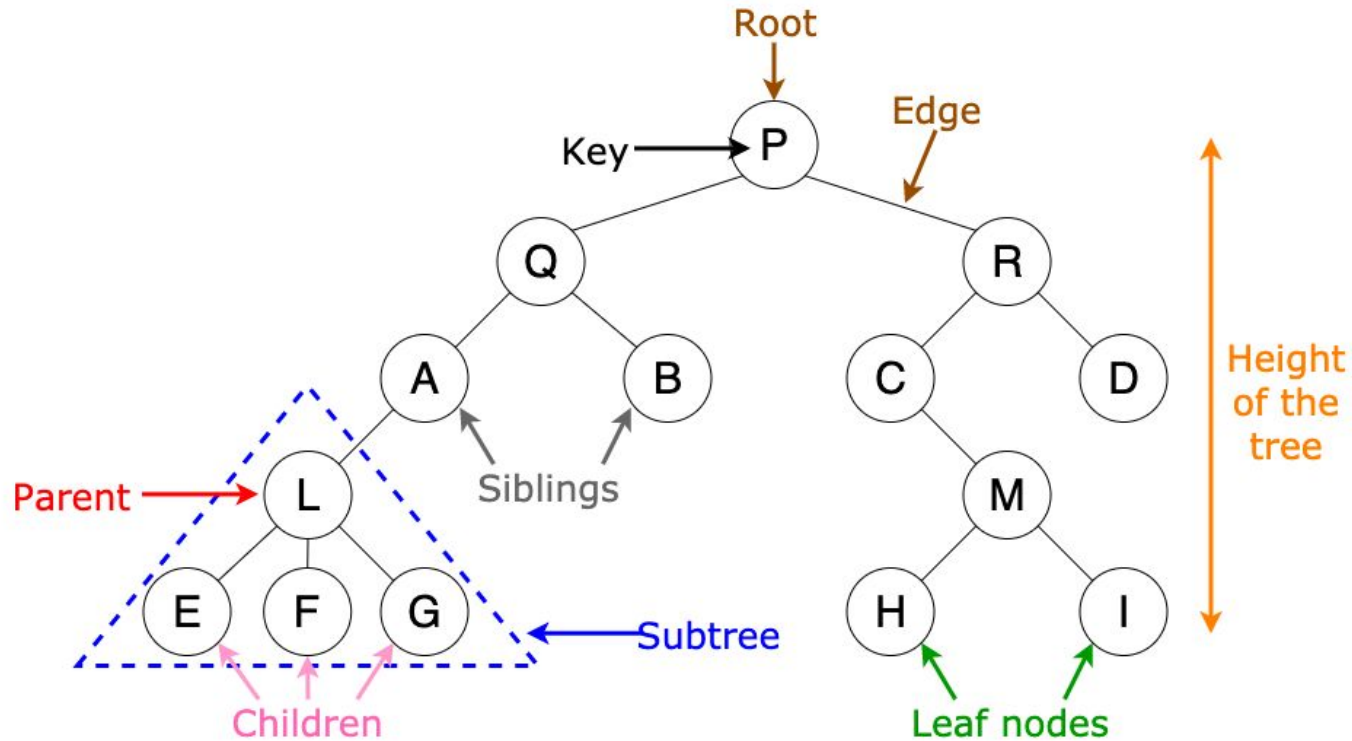
Trees - Root/Leaf/Non-Leaf



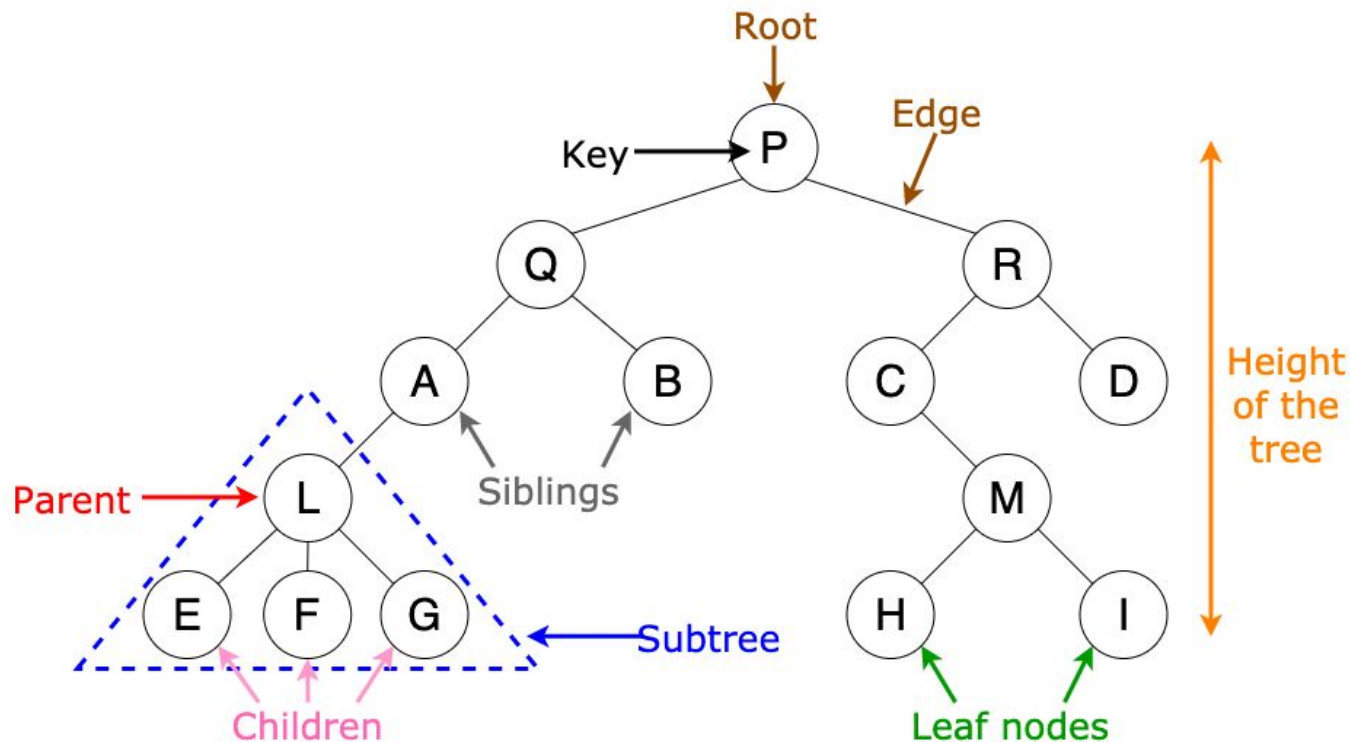
Trees - Parent/Child/Siblings



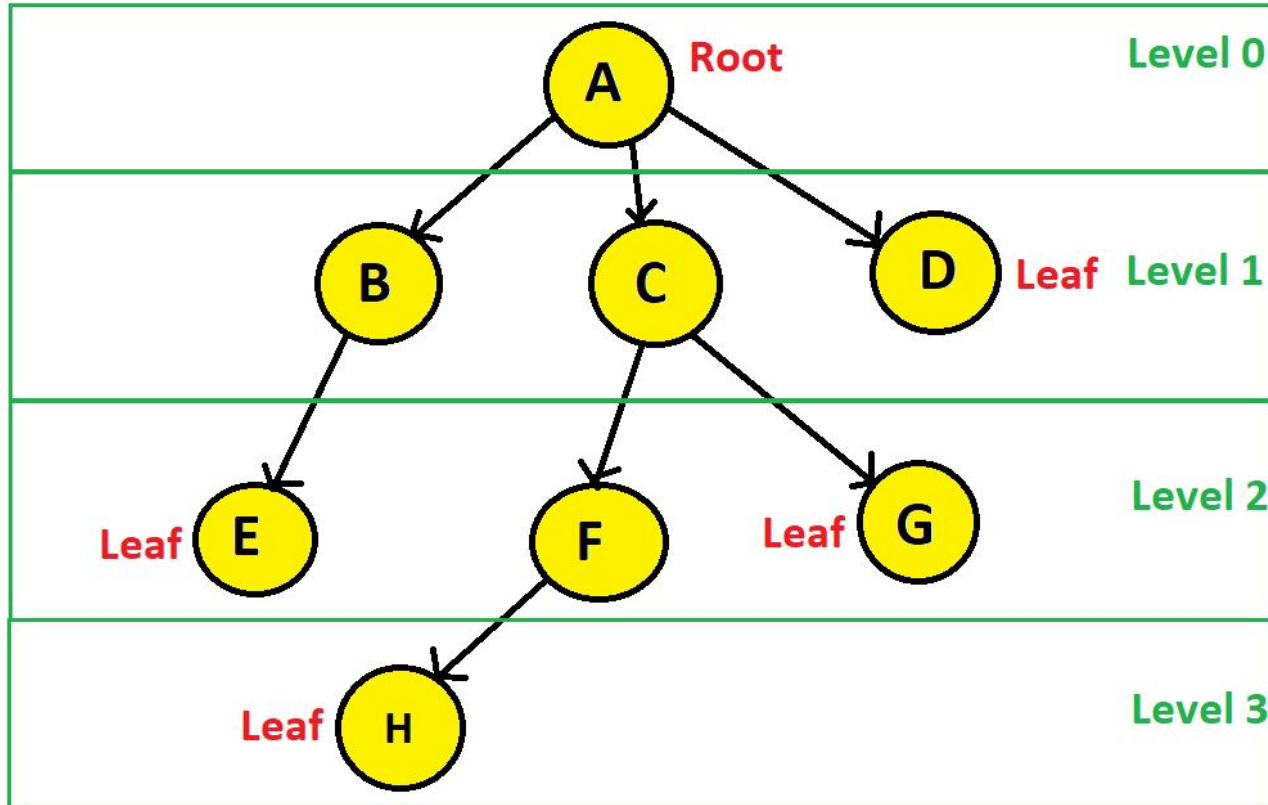
Trees - Edge/Path



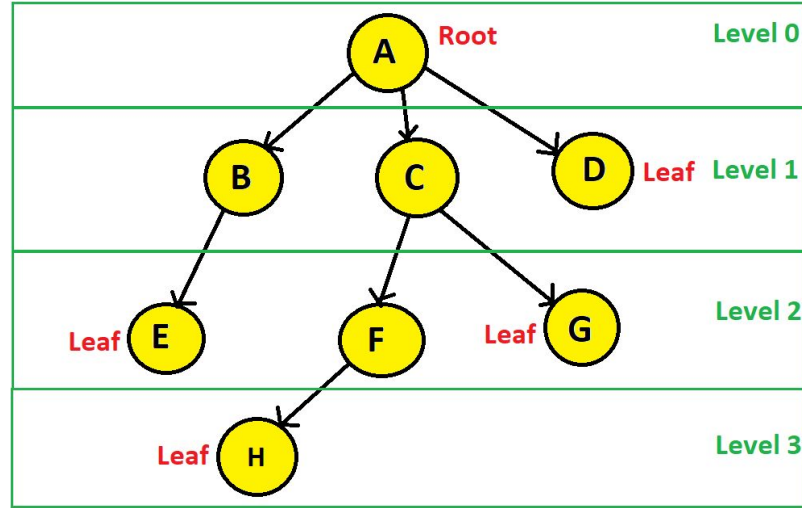
Trees - Degree



Trees - Depth/Height/Level



Trees - Depth/Height/Level



Depth of A?

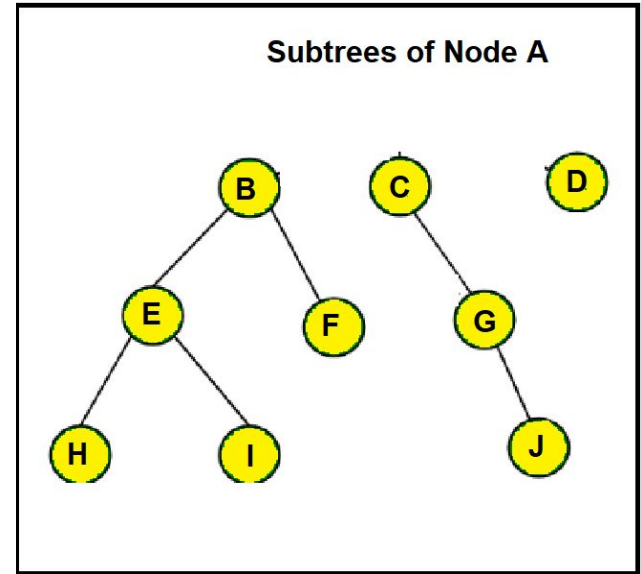
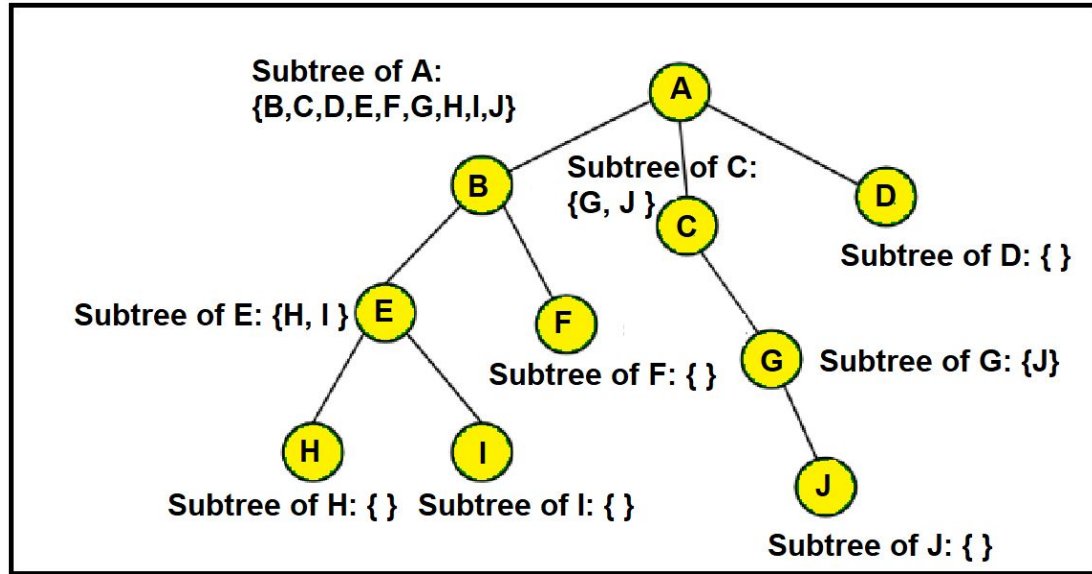
Height of

A?

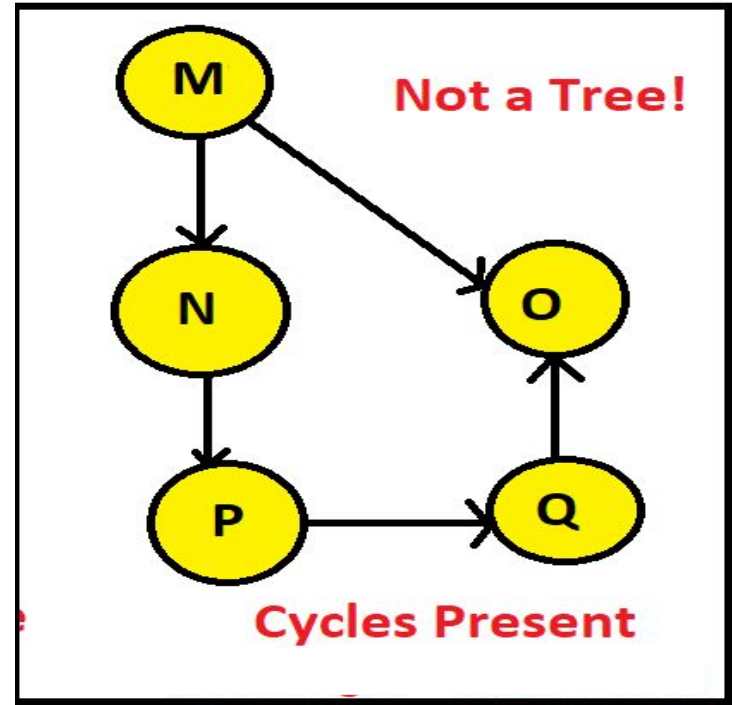
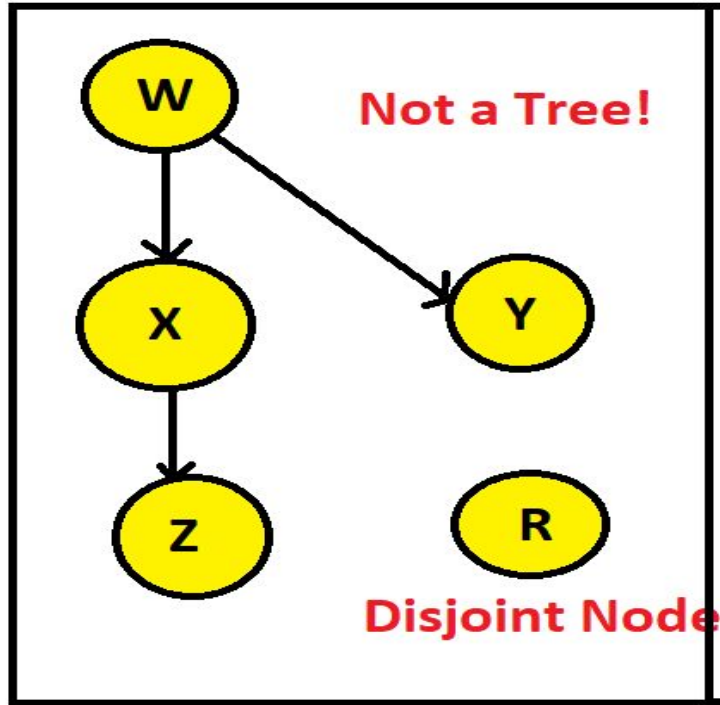
Depth = Level

Depth != Height

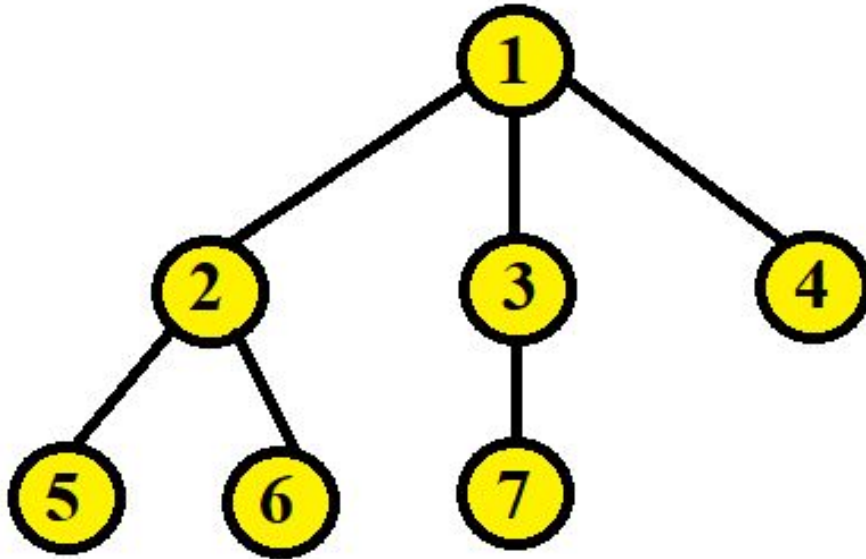
Trees - Subtree



Trees - Characteristics



Trees - Build a Tree



```
1 class Node:
2     def __init__(self, elem):
3         self.elem = elem
4         self.children = []
5
6 root = Node(1)
7 root.children += [Node(2)]
8 root.children += [Node(3)]
9 root.children += [Node(4)]
10 root.children[0].children += [Node(5)]
11 root.children[0].children += [Node(6)]
12 root.children[1].children += [Node(7)]
```

Diagram Representation with Adj List	Sequence Representation with Adj List
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